Capstone Project Concept Note and Implementation Plan

Project Title: [Your Project Title]

Team Members

1. [Team Member 1]
2. [Team Member 2]

Concept Note

1. Project Overview

- Provide a brief overview of your capstone project, emphasizing its relevance to one or more Sustainable Development Goals (SDGs).
- Clearly state the problem you aim to address and the potential impact of your solution.

2. Objectives

- Outline the specific objectives of your project.
- What do you aim to achieve, and how will the project contribute to addressing the identified problem?

3. Background

- Offer a comprehensive background that contextualizes the problem your project seeks to solve.
- Discuss any existing solutions or initiatives related to the issue, and explain why a machine learning approach is beneficial or necessary.

4. Methodology

- Explain the machine learning techniques and methodologies you plan to use.
- Highlight any specific algorithms, models, or frameworks that are crucial for the implementation of your project.

5. Architecture Design Diagram

- Provide a high-level overview of the architecture of your project.
    - Use a diagram to illustrate the key components and their interactions.
- Briefly describe each component shown in the diagram
    - Highlighting their roles and functionalities within the overall system.

6. Data Sources

- Detail the data sources you intend to use for your project.
- Include information about the type of data, its relevance to the problem, and any preprocessing steps required. **(You can use previous data review research but this should be 1 paragraph)**

7. Literature Review

- Summarize relevant literature that supports the methodology and approach you've chosen.
- Highlight key findings from existing research and explain how your project builds upon or extends this work. **(You can use previous literature review research but this should be 1 paragraph)**

Implementation Plan

1. Technology Stack

- List the technologies and tools you plan to use for the implementation of your project.
    - Including programming languages
    - Libraries
    - Frameworks
    - Any other software or hardware components.

2. Timeline

- Provide a detailed timeline for the different stages of your project,
    - Data collection and preprocessing
    - Model development
    - Training, and evaluation
    - Deployment
- Break down the timeline into manageable tasks with corresponding deadlines. You can use Gantt chart.
        - If you have pair you should add **task distribution matrix** to clarify each members responsibility.

3. Milestones

- Identify key milestones in your project's development.
    - These could be related to the completion of specific tasks, the achievement of certain performance metrics, or the successful implementation of key features.

4. Challenges and Mitigations

- Anticipate potential challenges that may arise during the project and propose strategies for mitigating them.
    - Data quality
    - Model performance
    - Technical constraints.

5. Ethical Considerations

- Discuss any ethical considerations associated with your project, especially concerning data privacy, bias, and the potential impact on the target community.

6. References